

Development of a wearable system with In-Ear EEG electrodes for the monitoring of brain activities: An application to epilepsy

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Abstract—Epilepsy is a chronic noncommunicable disease of the brain characterized by recurrent seizures, which are brief episodes of excessive and synchronized electrical discharges of neuronal activities. The main technique used for the diagnosis of seizures is the electroencephalography (EEG), a noninvasive method which records the electrical activity of the brain. The conventional EEG monitoring to diagnose epilepsy has some problems: First, the short monitoring time generates misdiagnosis. Second, the long wires needed for the connection of electrodes are uncomfortable for the patients. Third, the patients are not monitored in a natural environment where different conditions can trigger seizures and last, a conventional EEG monitoring is too expensive. A solution is the development of a wearable monitoring EEG where the recording system is a device of small dimensions with continuous and long-term recording capabilities. In this work, we developed an In-Ear wearable system to allow a long-term recording of EEG signals in a non-clinical environment. For this, we designed In-ear electrodes with soft foam earplugs and a knit conductive fabric of silver. We performed different tests to validate the signal quality of the acquisition system including one ambulatory recording in an epileptic patient.

Index Terms—Epilepsy, In-ear EEG Electrodes, Wearable System, Brain activity, Wireless, Long-term monitoring, Mobile application, Electroencephalography.

I. INTRODUCTION

Epilepsy is a disorder of the brain characterized by a durable predisposition to generate epileptic seizures. These seizures are transient occurrences of signs and symptoms due to abnormal excessive or synchronous neuronal activity [1]. There are about 50 and 100-190 cases of epilepsy per 100000 persons per year in developed and developing countries respectively [2], making it the most common neurological disorder.

The main technique used to diagnose epilepsy is the electroencephalography (EEG). The EEG is a noninvasive method which records the electrical activity of the brain [3]. Nevertheless, the conventional EEG monitoring for the epilepsy diagnosis has the following drawbacks: 1) The EEG monitoring in a clinical environment is performed with conventional scalp electrodes and it has a duration between 20 to 30 minutes.

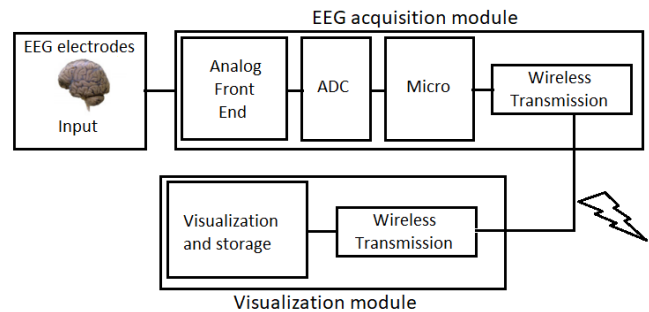


Fig. 1: Wearable EEG system.

Therefore, the short monitoring time generates misdiagnosis, and the long wires are uncomfortable for the patient [4]. 2) The patients are not monitored in a natural environment where different conditions can trigger seizures [4] and 3) A conventional EEG monitoring is too expensive [5].

In order to overcome these problems, a possible solution could be the development of wearable monitoring EEG where the recording system is a device of small dimensions with continuous and long-term and ambulatory recording capabilities [5]. For this, such a system should be lightweight, energy-efficient, and computationally powerful to acquire, process and communicate data [6]. Typical developments contain at least the following parts: An analog front end, an analog digital converter (ADC), a microcontroller, a type of wireless transmission (e.g. Bluetooth or WIFI), and a visualization and storage component (e.g. a mobile application or a PC) (see Fig. 1) [7], [8].

On the other hand, in recent years, a novel technique has emerged that reduces the size of wearable systems for the electrical signal monitoring of brain activities. This technology consists in the use of electrodes inside the ear canal with a small contact surface. Thus, the so-called in-ear EEG electrodes have a high signal quality, a good adaptability and a good compatibility [9], even compared to standard scalp electrodes as it has been previously shown [10], [11]. Furthermore, this type of electrodes is also able to record

epileptic activity, in particular, epileptic seizures from focal patients in controlled clinical environments [11].

In this work, we developed a wearable system with In-Ear electrodes for the continuous monitoring of EEG signals. In addition, we developed a validation protocol with healthy subjects and one epileptic patient.

II. MATERIALS AND METHODS

The study done was a pilot test, and subjects gave written an informed consent. The inclusion criteria were: the healthy subjects had the ability to do the activities mentioned and did not have any mental or physical illness. The epileptic patient had a controlled focal epilepsy and had not had a seizure in the last year. For minors, the informed consent was made by the parents. The developed wearable system is composed of the following parts: In-ear electrodes, an EEG acquisition device and a visualization module, as can be seen in Fig. 1.

A. In-ear EEG electrodes

In-ear electrodes were designed and developed using conductive fabric of silver, foam ear plugs, heat shrinkable and copper wire. We chose these materials because they had been already tested in previous studies (see Fig. 2) [12].

B. EEG acquisition module

For the EEG acquisition device, we used the OpenBCI Cyton Board that works with a sample rate of 250 Hz. This one contains a PIC32MX250F128B microcontroller with ChipKIT UDB32-MX2-DIP bootloader, an ADS1299 analog front end, a RFDuino BLE radio module, a Micro SD card slot and a DC battery of 3-6V. This board was designed to record EEG signals and send them to the computer or mobile via Bluetooth. For the communication to work, it was necessary to change the firmware of the board to have communication between the OpenBCI Cyton and a mobile application. The process of signal acquisition was as follows: First, we developed an algorithm for processing, sending and storage EEG signals in real time. The Fig. 3 shows the flowchart of the implemented algorithm in the PIC32MX250F128B. Additionally, a data storage was made in a microSD for offline analysis. Then, the module RFDuino was set up for the communication between the board and a mobile application. The wireless communication implemented was Bluetooth Low Energy (BLE).

C. Visualization module

We developed a mobile application (APP) in Android Studio for data visualization and control of the OpenBCI board. The Fig. 4 shows the flowchart we designed for the development of the APP.

D. Experimental Setups

Five tests were carried out to verify the correct operation of the system: First, we performed continuous EEG monitoring in a healthy subject for 5 minutes. The subject opened and closed his eyes at intervals of 30 seconds. EEG data was acquired simultaneously with In-ear electrodes and conventional gold cup electrodes at the scalp (Gold Cup 120 IN Philips

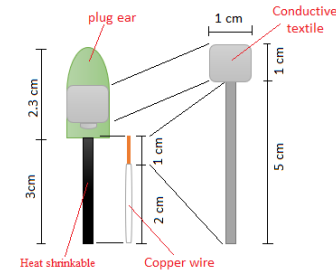


Fig. 2: Design and materials of In-ear EEG electrodes.

Table I: Block diagram of the tests made to verify the wearable system working

Subjects	Age	Condition	Electrodes	Recording time	Description
Subject 1	58	Control	In-ear EEG and gold cup electrodes	5 minutes	Open and close the eyes in intervals of 30sec
Subject 2	21	Control	In-ear EEG	10 minutes	Open and close the eyes in intervals of 30sec and 60 seconds respectively
				20 minutes	Play a puzzle game
				30 minutes	Watch TV
				24 hours	Do daily activities
Subject 3	53	Control	In-ear EEG	4 hours	Sleep
Subject 4	21	Control	In-ear EEG	25 minutes	Watch TV
Subject 5	9	Epileptic activity	In-ear EEG	1 hour	Sleep

Respironics). Second, we performed a longer continuous EEG monitoring in a healthy subject for 60 minutes. The goal of this test was to validate the recording of physiological EEG signals with the system. Third, we carried out a continuous EEG monitoring in a healthy subject for 60 minutes while the subject was doing daily activities (e.g. sleeping, walking, watching TV). Then, we performed an EEG monitoring in a healthy subject for 240 minutes while the subject was sleeping. Finally, we recorded EEG activities in an epileptic patient for 85 minutes while the patient was sleeping and watching TV. The complete validation protocol can be seen in Table I.

E. Offline Analysis

For the offline analysis, we implemented a bandpass filter between 3Hz and 80Hz, and a notch filter to eliminate the 60 HZ line noise. In addition, we computed the time frequency transform to identify the brain waves.

III. RESULTS

A. In-ear EEG electrodes

The Fig. 5 (a) shows the result of the designed electrodes. To reduce the noise, In-ear electrodes should be well connected to the ear canal. In some cases, we cut the electrode tip to assure that the conductive fabric did not contact the air.

B. Mobile Application (APP)

As can be seen in Fig. 6, the mobile application consists of three parts: The first part makes the Bluetooth configuration (Enabling, scanning and device's connection). The second part sets up the numbers of channels for EEG acquisition (between 1 and 8), the recording time to store in the SD (from 5 min to 24h) and the start/stop of EEG data reading. The third part

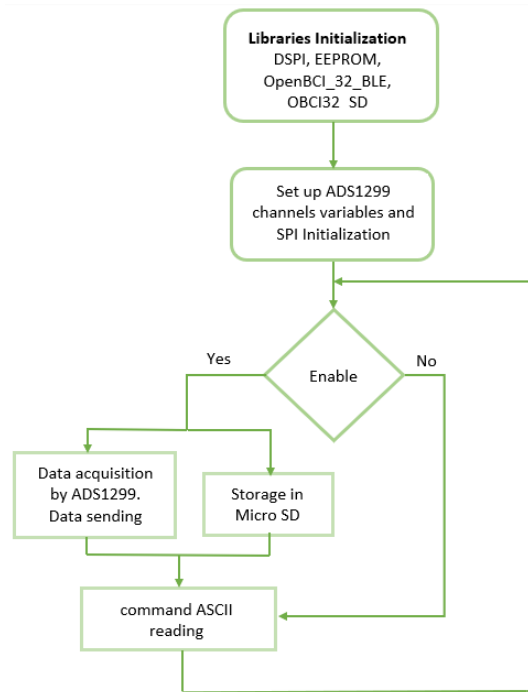


Fig. 3: Flowchart for processing, sending and storage of EEG signals.

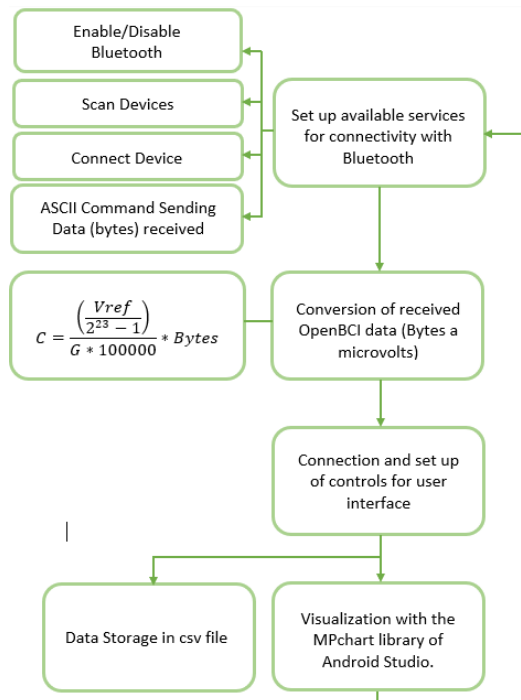


Fig. 4: Flowchart for data visualization in the APP. In the conversion of received data: C is the data in microvolts, V_{ref} is the reference voltage of the ADS1299, G is the gain of the ADS1299, and Bytes is the data in bytes.

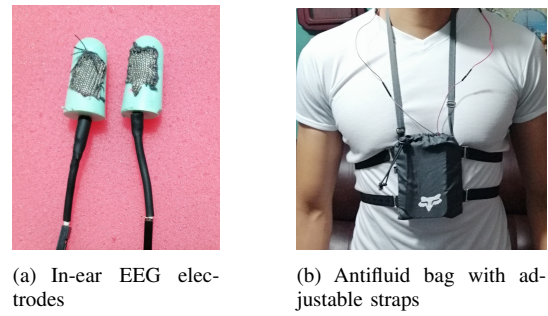


Fig. 5: Prototype of the wearable system



Fig. 6: Interface of the mobile application developed.

shows the visualization of the EEG signal in real time for channel 1.

C. Prototype of Wearable System

A first prototype was designed for the continuous monitoring tests. An antifluid bag was adapted with adjustable straps for different people sizes. Inside the bag there is a box with the OpenBCI Cyton Board and its packed batteries (see Fig. 5 (b)). The size of the wearable device is 12cm (L) x 8cm (h) x 3cm (w), and it weighs 130g.

D. Validation of stored EEG signals

We present EEG recordings from the five tests presented in the protocol (see Table I). The Fig. 7 shows the EEG signal recorded with ear-EEG electrodes and standard gold cup electrodes (test 1). The time frequency transform was used to confirm the occurrence of alpha activities (brain waves between 8 Hz and 14Hz) in the two types of electrodes. We calculated a cross-correlation value of 0.6483 between these two signals. In a subsequent test, we could identify alpha activity when the subject opened and closed his eyes (see Fig 8). However, it was not possible to identify significant neural oscillations when the subject watched TV and played a puzzle game. In the tests 3 and 4 while the subject was sleeping, we can see the presence of characteristic non Rapid Eye Movement slow waves (1Hz to 4 Hz) as shown in Fig. 9. As observed, the presence of the slow wave activity increased the EEG amplitude between 70uV to 100uV. In the last test, we recorded brain activity from an epileptic patient (Fig.

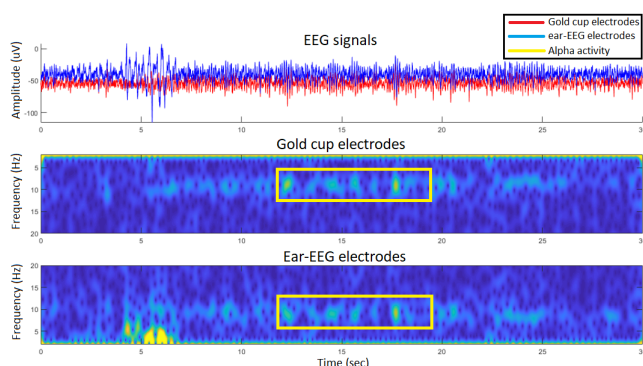


Fig. 7: Relation between EEG signals with ear-EEG electrodes and EEG signals with gold cup electrodes. This segment is from 210 seconds to 240 seconds.

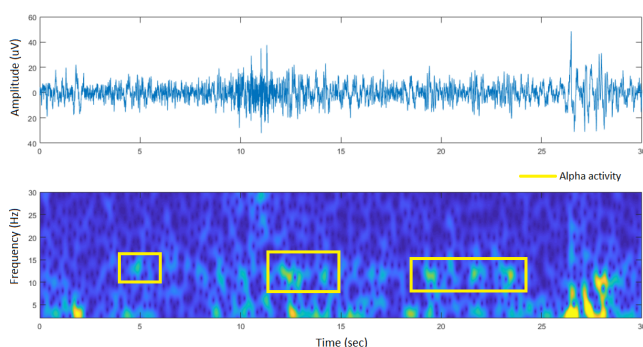


Fig. 8: EEG signal segment for 30 seconds in the interval from 540 seconds to 570 seconds. The task consists in opening and closed the eyes.

10). In these recordings, we could observe an epileptic-like activity (inter-ictal spikes), but nevertheless, a confirmation by a clinical expert was not performed and therefore it is required for future validations of the system. In addition, we checked the durability of the battery with a continuous recording. The total time of durability was 37 hours.

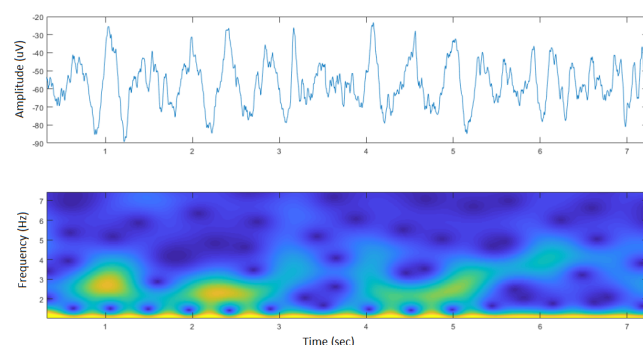


Fig. 9: Presence of slow waves while the person is sleeping. This segment is taken of the EEG signal acquired of the test 4.

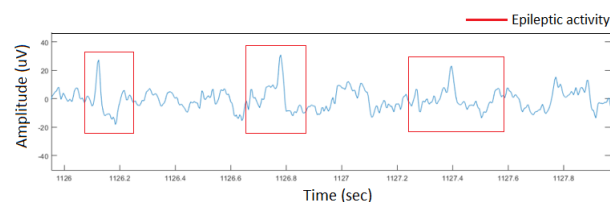


Fig. 10: EEG signal segment with presence of epileptic activity.

IV. CONCLUSION

Based on the results presented in this work, we think that the In-ear EEG electrodes provide a good alternative for the long term monitoring of brain signals in ambulatory contexts. In fact, the tests performed to validate the wearable system allowed us to identify different neural waves. In addition, the characteristics of the EEG signals with ear EEG electrodes was similar to the EEG signals with standard gold cup electrodes. Furthermore, it was possible to identify epileptic-like activity with a good signal quality. In the tests we performed, the subjects reported comfort with the electrodes; nevertheless, they manifested the reduction of sounds when extended uses. For instance, after having the electrodes connected for more than 8 hours, the subject began to feel unable to do some tasks. Concerning the portable bag, the subjects reported a good comfort due to the use of adjustable straps and the small weight of the device. In future work, the wearable system will be tested in more epileptic patients to better validate its capabilities for the identification of epileptic activities, including seizures, through the use of this type of electrodes.

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